## 11.9. Functions Defined by Power Series.

**Example:** The Bessel function of order 0 is defined by.

$$\mathcal{L}_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}.$$

a). Find the domain of $\mathcal{L}_0$.

b). Find the derivative of $\mathcal{L}_0$.

**Sln.** Let $a_n = \dfrac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^{n+1} x^{2n+2}}{2^{2n+2}((n+1)!)^2} \cdot \frac{2^{2n}(n!)^2}{(-1)^n x^{2n}}\right|$$

$$= \frac{x^2}{4} \frac{(n!)^2}{((n+1)!)^2}$$

$$= \frac{x^2}{4}\left(\frac{1}{n+1}\right)^2 \longrightarrow 0 \quad \text{as } n \to \infty.$$

So, $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 0 < 1$ then $\text{Dom}\,\mathcal{L}_0 = \mathbb{R}$.

b) $$\frac{d}{dx}(\mathcal{L}_0(x)) = \frac{d}{dx}\left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}\right)$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2n\, x^{2n-1}}{2^{2n}(n!)^2}$$

## 11.10 Taylor and Maclaurin Series.

**Def:** Let consider a function such that

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n. \quad (***)$$

this is a Taylor Series of $f(x)$ centered at $x = a$.

if $a = 0$ then $(***) = \sum_{n=0}^{\infty} c_n x^n$ is called a Maclaurin Series.

**Thm:** If $f$ has a power series representation (expansion) at $a$, i.e.,

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n \qquad |x-a| < R$$

then its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}$$

**Rmk:**

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots$$

**Example:** We know that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

let $f(x) = \frac{1}{1-x}$ ; $f(0) = 1$.

$f'(x) = \frac{1}{(1-x)^2}$ ; $f'(0) = 1$

$f''(x) = \frac{1 \cdot 2}{(1-x)^3}$ ; $f''(0) = 1 \cdot 2$

$f'''(x) = \frac{1 \cdot 2 \cdot 3}{(1-x)^4}$ ; $f'''(0) = 1 \cdot 2 \cdot 3$

$\vdots$

$f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$ ; $f^{(n)}(0) = n!$

$$c_n = \frac{f^{(n)}(0)}{n!} = \frac{n!}{n!} = 1.$$

**Example:** For the function $f(x) = e^x$, find the Maclaurin series and its radius of convergence.

**Sln:** $f(x) = e^x$ $\qquad f(0) = 1$.

$f'(x) = e^x \qquad f'(0) = 1$

$f''(x) = e^x \qquad f''(0) = 1$

$\vdots$

$f^{(n)}(x) = e^x \qquad f^{(n)}(0) = 1$

then

$$e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

To find the radius of convergence, $a_n = \frac{x^n}{n!}$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \frac{|x|}{n+1} \longrightarrow 0 < 1$$

then the radius is $R = \infty$.

**Example.** Find the Taylor Series for $f(x) = e^x$ at $a = 2$

**Sln.**

$f(x) = e^x \Rightarrow f(2) = e^2$

$f'(x) = e^x \Rightarrow f'(2) = e^2$

$\vdots$

$f^{(n)}(x) = e^x \Rightarrow f^{(n)}(2) = e^2$

then,

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!} (x-2)^n = \sum_{n=0}^{\infty} \frac{e^2}{n!} (x-2)^n$$

$$a_n = \frac{e^2}{n!} (x-2)^n$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{e^2 (x-2)^{n+1}}{(n+1)!} \cdot \frac{n!}{e^2 (x-2)^n} \right| = \frac{|x-2|}{n+1} \xrightarrow[n \to \infty]{\text{as}} 0$$

then $R = \infty$.

## Taylor Series of important functions.

1). Find the Maclaurin Series for $\sin x$ and prove that it represents $\sin x$ for all $x$.

**Sln:**

$f(x) = \sin x \qquad f(0) = 0.$

$f'(x) = \cos x \qquad f'(0) = 1 \qquad f^{(2n+1)}$

$f''(x) = -\sin x \qquad f''(0) = 0.$

$f'''(x) = -\cos x \qquad f'''(0) = -1.$

$f^{(4)}(x) = \sin x \qquad f^{(4)}(0) = 0.$

Since the derivatives repeat in a cycle of four, we can write:

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!} + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$\begin{cases} \text{if } n = 2k \Rightarrow f^{(2k)}(0) = 0 \\ \text{if } n = 2k+1 \Rightarrow f^{(2k+1)}(0) \neq 0. \end{cases}$$

$$= \sum_{k=0}^{\infty} \frac{f^{(2k+1)}(0)}{(2k+1)!} x^{2k+1}$$

Notice that $f^{(2k+1)}(0) = (-1)^k$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}.$$

Let $a_k = \dfrac{(-1)^k x^{2k+1}}{(2k+1)!}$

$$\left| \frac{a_{k+1}}{a_k} \right| = \left| \frac{(-1)^{k+1} x^{2k+3}}{(2k+3)!} \cdot \frac{(2k+1)!}{(-1)^k x^{2k+1}} \right|$$

$$= x^2 \frac{1}{(2k+2)(2k+3)} \xrightarrow[k \to \infty]{\text{as}} 0$$

So, $R = \infty$.

**Example:** Find the Maclaurin series for $\cos x$.

$$\cos x = \frac{d}{dx}(\sin x) = \frac{d}{dx}\left( \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \right)$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

**Example:** Represent $\sin x$ as the sum of its Taylor series centered at $x = \frac{\pi}{3}$.

**Sln:**

$$f(x) = \sin x \qquad f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$f'(x) = \cos x \qquad f'\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$f''(x) = -\sin x \qquad f''\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$f'''(x) = -\cos x \qquad f'''\left(\frac{\pi}{3}\right) = -\frac{1}{2}$$

$$f\left(\frac{\pi}{3}\right) + \frac{f'\left(\frac{\pi}{3}\right)}{1!}\left(x - \frac{\pi}{3}\right) + \frac{f''\left(\frac{\pi}{3}\right)}{2!}\left(x - \frac{\pi}{3}\right)^2 + \cdots$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{2 \cdot 1!}\left(x - \frac{\pi}{3}\right) - \frac{\sqrt{3}}{2 \cdot 2!}\left(x - \frac{\pi}{3}\right)^2 - \frac{1}{2 \cdot 3!}\left(x - \frac{\pi}{3}\right)^3 + \cdots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2\,(2n+1)!}\left(x - \frac{\pi}{3}\right)^{2n+1} + \sum_{n=0}^{\infty} \frac{(-1)^n \sqrt{3}}{2\,(2n)!}\left(x - \frac{\pi}{3}\right)^{2n}$$

$f^{(2n+1)}\left(\frac{\pi}{3}\right) = (-1)^n \cdot \frac{1}{2}$

$f^{(2n)}\left(\frac{\pi}{3}\right) = (-1)^n \frac{\sqrt{3}}{2}$.